Untitled

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R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```r
# load libraries
library(vroom)
library(here)
```

```
## here() starts at /home/ab/Dropbox/university/github/BIOLM0039_exam
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.2.0
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x readr::col_character()   masks vroom::col_character()
## x readr::col_date()        masks vroom::col_date()
## x readr::col_datetime()    masks vroom::col_datetime()
## x readr::col_double()      masks vroom::col_double()
## x readr::col_factor()      masks vroom::col_factor()
## x readr::col_guess()       masks vroom::col_guess()
## x readr::col_integer()     masks vroom::col_integer()
## x readr::col_logical()     masks vroom::col_logical()
## x readr::col_number()      masks vroom::col_number()
## x readr::col_skip()        masks vroom::col_skip()
## x readr::col_time()        masks vroom::col_time()
## x readr::cols()            masks vroom::cols()
## x readr::date_names_lang() masks vroom::date_names_lang()
## x readr::default_locale()  masks vroom::default_locale()
## x dplyr::filter()          masks stats::filter()
## x readr::fwf_cols()        masks vroom::fwf_cols()
## x readr::fwf_empty()       masks vroom::fwf_empty()
## x readr::fwf_positions()   masks vroom::fwf_positions()
## x readr::fwf_widths()      masks vroom::fwf_widths()
## x dplyr::lag()             masks stats::lag()
## x readr::locale()          masks vroom::locale()
```

```
## x readr::output_column()   masks vroom::output_column()
## x readr::problems()        masks vroom::problems()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(Hmisc)
```

```
##
## Attaching package: 'Hmisc'
##
## The following objects are masked from 'package:dplyr':
##
##     src, summarize
##
## The following objects are masked from 'package:base':
##
##     format.pval, units
```

```r
library(ggpubr)
library(rstatix)
```

```
##
## Attaching package: 'rstatix'
##
## The following object is masked from 'package:stats':
##
##     filter
```

```r
library(fitdistrplus)
```

```
## Loading required package: MASS
##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:rstatix':
##
##     select
##
## The following object is masked from 'package:dplyr':
##
##     select
##
## Loading required package: survival
```

```r
library(MASS)
library(performance)
library(see)
library(TMB)
library(glmmTMB)
library(DHARMa)
```

```
## This is DHARMa 0.4.7. For overview type '?DHARMa'. For recent changes, type news(package = 'DHARMa')
```

Including Plots

You can also embed plots, for example:

```
## Rows: 870 Columns: 10
## -- Column specification --------------------------------------------------------
```

```
## Delimiter: "\t"
## chr (8): student_id, species, park, observer, landscape, vegetation, lion_pr...
## dbl (2): body_mass, env_value
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
## df
##
##  10  Variables      870  Observations
## --------------------------------------------------------------------------------
## student_id
##        n  missing distinct    value
##      870        0        1  sj19031
##
## Value      sj19031
## Frequency      870
## Proportion       1
## --------------------------------------------------------------------------------
## species
##        n  missing distinct    value
##      870        0        1  Giraffe
##
## Value      Giraffe
## Frequency      870
## Proportion       1
## --------------------------------------------------------------------------------
## park
##        n  missing distinct
##      870        0        4
##
## Value           Golden Tusk Safari    Lionshade Wilderness
## Frequency                      264                     186
## Proportion                   0.303                   0.214
##
## Value      Savanna Whisper Reserve      Zebra Horizon Park
## Frequency                      189                     231
## Proportion                   0.217                   0.266
## --------------------------------------------------------------------------------
## observer
##        n  missing distinct
##      870        0        5
##
## Value      Obs_1 Obs_2 Obs_3 Obs_4 Obs_5
## Frequency    147   204   159   183   177
## Proportion 0.169 0.234 0.183 0.210 0.203
## --------------------------------------------------------------------------------
## body_mass
##        n  missing distinct     Info     Mean  pMedian      Gmd      .05
##      870        0      290        1     1271     1233    383.9    844.2
##      .10      .25      .50      .75      .90      .95
##    889.9   1009.1   1212.1   1456.5   1673.3   1799.4
##
## lowest : 587.917 665.151 681.379 690.192 690.787
```

```
## highest: 2404.45 2775.11 3319.15 3503.69 4682.4
## ---------------------------------------------------------------------------------------
## landscape
##        n  missing distinct
##      861        9        2
##
## Value       Flat Hilly
## Frequency    441   420
## Proportion 0.512 0.488
## ---------------------------------------------------------------------------------------
## vegetation
##        n  missing distinct
##      861        9        2
##
## Value      Dense  Open
## Frequency    426   435
## Proportion 0.495 0.505
## ---------------------------------------------------------------------------------------
## lion_presence
##        n  missing distinct
##      861        9        5
##
## Value       Absent  Asbent Preesnt Preseet Present
## Frequency      405       3       3       3     447
## Proportion   0.470   0.003   0.003   0.003   0.519
## ---------------------------------------------------------------------------------------
## env_var
##        n  missing distinct
##      870        0        3
##
## Value      average_annual_rainfall_mm       distance_to_water_km
## Frequency                         290                        290
## Proportion                      0.333                      0.333
##
## Value      predator_density_per_sq_km
## Frequency                         290
## Proportion                      0.333
## ---------------------------------------------------------------------------------------
## env_value
##         n   missing  distinct      Info      Mean   pMedian       Gmd       .05
##       862         8       862         1     334.9     317.5     481.3    0.4668
##       .10       .25       .50       .75       .90       .95
##    0.6515    1.2368    3.5664  795.3547 1161.8979 1290.3489
##
## lowest : 0.101891 0.167874 0.172813 0.228129 0.259863
## highest: 1712.42  1714.88  1729.07  1737.42  1807.51
## ---------------------------------------------------------------------------------------
```

Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.